

Memory-based Personality Robot for Older Adults

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Abstract—With societal changes and the backdrop of an increasing proportion of aging and solitary living elders, more and more older people are facing the problem of social disconnection. Personalized robots are seen as a solution to promote social connections. The purpose of this study is to design and test the interaction between personalized robots and the older adults. The design method for the robot's personality uses Myers-Briggs Type Indicator (MBTI), and the technology involves a memory-based personality agent and robot system. The research method compares the differences between two different language models, gpt-4o and gpt-4-turbo, and observes the interaction between the older adults and robots using the gpt-4-turbo language model. Finally, the study discusses and suggests improvements for robot design based on the observation results.

I. INTRODUCTION

As time progresses, advancements in medical, technology and public health awareness have led to a gradual increase in the aging population in Taiwan. Taiwan entered an aging society in 1993 and is expected to become a super-aged society by 2026[1]. With the increase in the elderly population, the issue of aging requires more attention.

Furthermore, aging has also led to changes in the traditional family structure. In Taiwan, the number of multigenerational households (e.g. grandparent-grandchild and three-generation families) is decreasing, while the proportion of elderly living alone is increasing[1]. Due to living alone and facing significant life transitions such as retirement and the loss of a spouse, the elderly are at a higher risk of social disconnection. When people lose their connections, it further contributes to an internal sense of loneliness. Loneliness is subjectively perceived as a gap between actual social relationships and personal expectations, while objectively, it refers to a state of social isolation and lack of social connections[2][3].

Previous research has found that the negative subjective and objective impacts of loneliness will affect the physical and psychological functioning of the elderly[2][4]. Therefore, through psychosocial care and support, reducing the elderly's

perceived loneliness and promoting protective factors for health are essential areas of focus for community services for the elderly.

In contrast to loneliness, social connectedness refers to the subjective experience of belonging to social relationships or networks[5]. It encompasses various indicators of fulfilling the need for connection, such as a sense of belonging, acceptance, and social support[6]. According to previous research[7], social connectedness and loneliness are negatively correlated, indicating that they are two sides of the same coin. Therefore, through intervention, the passive goal is to reduce the elderly's loneliness, while the active goal is to promote their social connectedness. The study will focus on the positive aspect, aiming to enhance the elderly's social connectedness through intervention, thereby improving their quality of life.

The elderly are one of the primary target audiences for the development of social robots, and this study aims to enhance their sense of social connectedness. Previous research[8] has surveyed the needs of the elderly and their families regarding social robots. The top three functions desired by the elderly were: (1) assistance with daily activities, such as walking; (2) engaging in personalized interactive activities; (3) maintaining social connections. The top three functions deemed important by family members were: (1) reminder functions; (2) monitoring the health and safety of the elderly; (3) maintaining social connections. The survey results indicate that both the elderly and their families recognize the need for maintaining social connectedness[9].

For individuals with social connection needs, previous research[10][11] has found that they tend to anthropomorphize non-human agents (robots) to fill social gaps or as a way to explore their social world or expand their social network[12]. Studies[13] on cultural factors have found that in East Asian cultures, people are more inclined to view robots as humans, using them as an active means to expand social connections and networks, particularly for those lacking social connections. Therefore, East Asian cultures are more likely to support a growth-oriented approach to anthropomorphism, viewing the anthropomorphization of non-human agents as a genuine interest in these agents and considering them a part of their social world.

~~Resrach supported by ABC Foundation.~~

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Therefore, this study will target elderly with social connection needs. These individuals tend to anthropomorphize robots during interactions. Through anthropomorphic design, we aim to create robots with personalities and personal background stories, which we expect to have the potential to serve as social partners for the elderly.

II. BACKGROUNDS

Personality has been identified as a crucial factor in understanding the quality of robot deployments in organizations and in broader society[14]. Although the uncanny valley hypothesis posits that humans feel uncomfortable in the presence of robots with human-like features, this unsettling first impression is significantly altered through interaction[15]. The personality of a robot is considered preferable and is associated with desirable social responses[16]. To determine a suitable personality for a robot, prior research suggests that preferences for robot personalities may indeed vary depending on the context of the robot's role and the stereotypical perceptions people hold for certain occupations[17]. It has been demonstrated that users are capable of distinguishing between robot personalities, which results in differing preferences between goal-oriented and experience-oriented scenarios[18].

A robot has been designed to exhibit traits of introversion and extroversion to assist post-stroke individuals in their rehabilitation exercises[19]. Research indicates that a socially assistive robot's autonomous behavior, when adapted to the user's personality, can enhance task performance. Studies have demonstrated that a robot's extroversion and dominance influence people's perceptions of its intelligence, social capabilities, and likability[20]. Furthermore, the robot's personality has been shown to improve human-robot interactions; users report greater enjoyment when interacting with a robot whose personality complements their own, as opposed to one with a similar personality[21]. Guidelines for the effective design of service robots have been proposed to elicit desired emotional responses from users[22]. The incorporation of personality into robots facilitates human-robot interactions with greater social presence, leading to outcomes such as increased acceptance and heightened emotional engagement during service encounters. Previous research has primarily employed the Big Five model to instill personality in robots [23]. In contrast, interpersonal theory emphasizes interactions with others rather than internal characteristics[24], diverging from the Big Five model. Moreover, the assessment of robot personality has been calibrated using the International Personality Item Pool (IPIP) and the Big Five Inventory (BFI). However, these approaches are limited to the five dimensions of the Big Five Model and tend to concentrate solely on conversational aspects, neglecting the underlying thoughts and decisions. Consequently, it is imperative to develop a personality model for robots that can emulate cognitive processes and personality inference.

III. METHODS AND MATERIALS

A. MBTI robot personality design

Robert et al. (2020) reviewed 25 studies on robot personality, with each study utilizing design and experimental methods. Typically, researchers begin with a literature review

to confirm components of human personality, which are then applied to robot personalities. The most common approach involves developing robot personalities based on the NEO Personality Inventory (NEO-PI-R) (Costa & McCrae, 1992).

Secondly, researchers investigate the elements necessary to construct robot personalities (Hendriks et al., 2011). Third, they proceed with a development phase at which prototypes reflecting these traits are created. Finally, experiments are conducted to identify whether the robots are perceived to possess the expected traits and to explore the effects of these personalities.

The procedure for setting robot personalities is as follows:

1. Literature Review: Relevant literature and studies are referenced to understand which robots' personality traits make humans more willing to interact with the robots.
2. Identification of Personality Traits: Based on the literature, people are more willing to build social connections with those who have traits such as vulnerability, emotional sensitivity, indecisiveness, need for dependence on others, and willingness for self-disclosure (Martelaro et al., 2016).
3. MBTI Testing: Characters are established according to these traits, and MBTI questions are answered based on these character roles to determine the MBTI type.
4. Confirmation of Robot Personality: After responses from 3 out of 5 team members, the team measured the robot's MBTI type as ENFJ and used the description of the ENFJ personality as the foundation for designing the robot prototype.

B. Memory-based Personality Agent

An memory-based personality agent is designed with the concept of personality and memory. As shown in Figure 1, the architecture of memory-based personality agent consists of three parts: memory function, personality inference and intent loader. The main objective of the design is to realize the personality design with memory and intent.

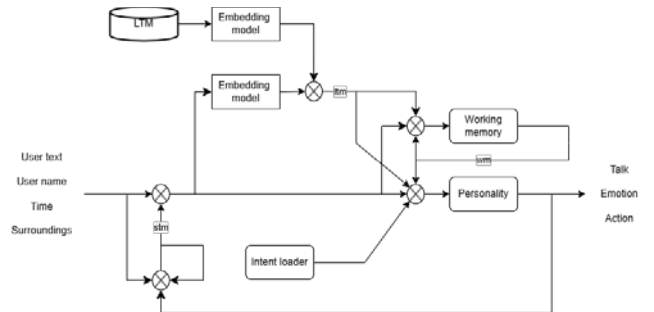


Figure 1. Architecture of memory-based personality agent

Derived from social cognitive theory, the state of environment and agent at turn k are $X_{environment}$ and X_{agent} and the outputs are $Y_{environment}$ and Y_{agent} described by (1)~(4). There are infinite states environment according to the representation, such that the states of environment mainly indicate the input from user who is talking to the robot in this research.

$$X_{agent}[k] = \begin{bmatrix} x_{talk} \\ x_{action} \\ x_{emotion} \\ x_{stm} \\ x_{wm} \\ x_{ltm} \end{bmatrix} [k] \quad (1)$$

$$Y_{agent}[k] = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} X_{agent}[k] \quad (2)$$

$$X_{environment}[k] = \begin{bmatrix} x_{user\ text} \\ x_{user\ name} \\ x_{time} \\ x_{surroundings} \\ \vdots \end{bmatrix} [k] \quad (3)$$

$$Y_{environment}[k] = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ \vdots \end{bmatrix} X_{environment}[k] \quad (4)$$

The interaction loop between agent and environment becomes:

$$= f_{agent}(Y_{environment}[k], X_{agent}[k-1]) \quad (5)$$

$$= f_{environment}(Y_{agent}[k], X_{environment}[k]) \quad (6)$$

where the f_{agent} is regarded as the robot and $f_{environment}$ is regarded as the user.

The inference module implement the personality design and infer a suitable output according to user input. The inference considers the user input, intent of robot, short-term memory, long-term memory and working memory as shown in (7).

$$Y_{agent}[k] = f_{inference}(Y_{environment}[k], x_{intent}[k], x_{stm}[k], x_{ltm}[k], x_{wm}[k]) \quad (7)$$

The memory function of memory-based personality robot architecture includes three types of memory: long-term memory, short-term memory and working memory. The short-term memory records 7 turns of conversation, for the working memory capacity of humans is about 7+-2. To maintain some important information over 7 turns, the working memory model automatically selects the information related to **rules and requirements**. It enables the robot to remember the user's name after the first mutual introduction.

$$= f_{wm}(Y_{environment}[k], x_{stm}[k], x_{ltm}[k], x_{wm}[k-1]) \quad (8)$$

On the other hand, long-term memory enables the robot to retrieve the conversation history from a large database, eliminating the restrictions of data size. The retrieval cue is transformed into a vector, which is then used to search for related memory vectors employing the Maximal Marginal Relevance (MMR) algorithm [1]. The query and text are embedded into vectors with 512 dimensions:

$$v[k]_{1 \times 512} = V\{x[k]\} \quad (9)$$

MMR finds out the similar content between standalone statement and Long-term memory (L) by considering both diversity and relativity.

$$c \in \left\{ \text{Arg} \max_{d_i \in L} \left[\lambda \text{sim}(v_{standalone}[k], V\{d_i\}) - (1 - \lambda) \max_{d_j} (\text{sim}(V\{d_i\}, V\{d_j\})) \right] \right\} \quad (10)$$

Outperforming the Euclidean Distance method, MMR strikes a balance between relevance and diversity, thereby providing comprehensive information for formulating responses to the user. Additionally, memory retrieval by timing labels is conducted through the optimization of character string comparisons to ensure maximal satisfaction. With the design of the memory modules, the robot can recall necessary information from different types of memory, performing a human-like character with high flexibility of conversation.

The intent loader leads the overall conversation based on the autobiography interview. Four stages are implemented to facilitate the conversation and build up the relationship between robot and user, and each of the stages includes two objectives, as shown in Figure 2. By changing the stage with turns of conversation, the robot can progress the conversation and lead the interaction with user.

Intent-driven conversation

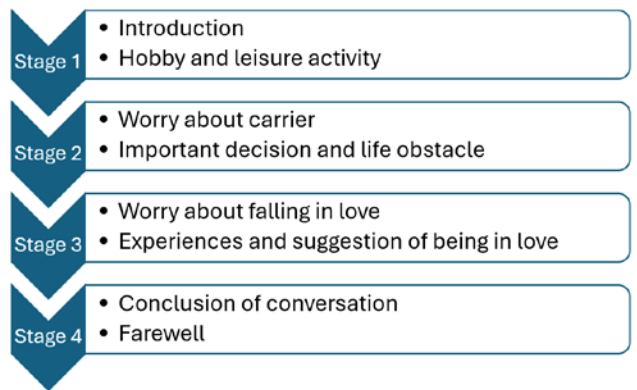


Figure 2. 4 stages of objective in intent-driven conversation

C. Robot system

Following the detailed methodology for designing the robot's personality and conversational capabilities, this

section introduces the complete robot system, which is divided into a Python-based server and a Java-based client. These components communicate through a socket protocol to facilitate data exchange. This system is implemented on a commercial robot from Nuwa Robotics, known as Kebbi, which comes equipped with specific hardware designed to support its interactive functionalities.

The robot system is built to leverage the strengths of both Python and Java, ensuring efficient management of conversational data and physical interactions. The Python server is responsible for generating personalized dialogues using advanced language models, while the Java client manages speech recognition (ASR), message transmission, text-to-speech (TTS), and the execution of the robot's physical actions and expressions.

- **Python Server:**

Running on the Python server, the memory-based personality agent is implemented as the core of the entire robot system by using Large Language Model (LLM) and Retrieval-Augmented Generation (RAG), as mentioned in III B, which mainly infers the reply, emotion and action of the robot based on the input from the user.

- **Java Client:**

The Java client is responsible for handling the robot's physical interactions. This includes speech recognition (ASR), text-to-speech (TTS), message transmission and reception, and control of the robot's movements and expressions. The Java client utilizes the Nuwa SDK for these functions, ensuring seamless integration and optimal performance.

- **Speech Recognition (ASR):**

The ASR module captures and transcribes spoken language into text, enabling the robot to understand and process user inputs. This functionality is essential for initiating and maintaining conversations.

- **Text-to-Speech (TTS):**

Converts text responses generated in Java with Nuwa SDK into natural-sounding speech, allowing the robot to verbally communicate with users. This ensures that the robot's responses are clear and engaging.

- **Message Transmission and Reception:**

The Java client sends user inputs to the Python server and receives generated responses. This real-time data exchange is facilitated by the socket protocol, ensuring smooth interaction.

- **Movements and Expressions:**

The robot's movements and facial expressions are controlled using the Nuwa SDK, which provides predefined animations and gestures to enhance the interactive experience, allowing the user to understand the current operating state of the robot, which is a crucial design principle in the realm of HRI [2]. These

physical responses are synchronized with the conversation to make interactions more lifelike.

The entire system is implemented on Nuwa Robotics' commercial robot, Kebbi, featuring hardware with 12 degrees of freedom, 5 sensing areas and an Android tablet. The tablet is running on Android 9, KebbiOS 11.2. As shown in Figure 3, Kebbi's hardware includes cameras for visual input, microphones for audio input, and speakers for audio output, all of which are essential for its conversational functionalities.

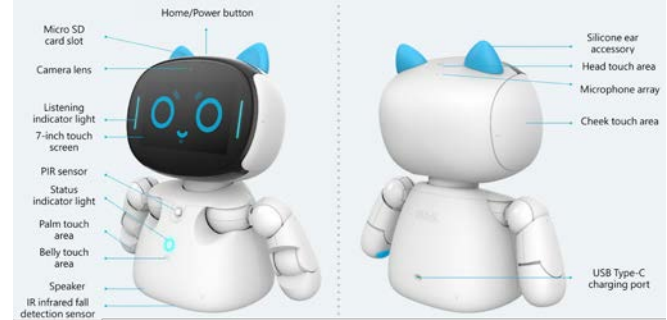


Figure 3. 4 stages of objective in intent-driven conversation

IV. CONCLUSION

A. Differences Between the GPT-4O Model and the 4O Model

In comparison to robots implementing the GPT-4O model, robots employing the GPT-4TURBO model exhibit more lively speech responses. For example, they may say phrases like "Sounds very relaxing" and "The method may be old-fashioned but quite effective." Moreover, this model extends conversation topics during chats, such as asking the interlocutor: "Do you prefer walking alone or with friends?" and "Is following the family business a good choice, or should one consider forging their own path?" Lastly, responses from this model demonstrate a warm and empathetic personality, exemplified by statements like "Repeatedly thank the other person for their affirmation and assistance, expressing hopes of becoming a better person" and "Early career was quite challenging." These characteristics align more closely with our original intent in designing personalized companion robots.

Compared to the GPT-4O model, the GPT-4TURBO model has a longer response time, extended by two seconds. The 4O model performs better in terms of response speed.

B. The interaction between the elderly and the robot with the GPT-4TURBO model

Some elderly individuals felt quite nervous when speaking to a robot for the first time, fearing that the robot may not understand what they say or the operating of robots could be challenging. However, others interacted with robots positively and enthusiastically. They found robots fascinating and were eager to assist in improving them. They feel valued in contributing their conversation styles, content, pauses, intonations, and other aspects to optimize interactions with robots.

In the initial rounds of conversation, because the robot's appearance resembled a child, the participants treated the robot as a child, creating a top-down relationship. The participants

did not perceive the robot as the intended university student role. However, as the human-robot interaction progressed to the mid and later stages, it became evident that the robot's dialogue successfully portrayed the personality of a university student. Consequently, the participants adjusted their manner of speaking, treating the robot as a university student. This indicates that the social role was successfully conveyed, convincing users to modify their tone and content, and continue engaging in conversation with the robot.

V. DISCUSSION AND FUTURE OUTLOOK

The robot can have a smooth conversation with the older adults, as shown in Figure 4. The feedback gathered from the end-of-term report highlights several critical aspects that need to be addressed to enhance the robot's interaction design. One significant issue is the long waiting time for users to receive responses, which needs to be mitigated. To improve the conversational flow, the robot should acknowledge the user's statements with empathic feedback and then pose questions at different levels of complexity. For instance, suggested conversational strategies include providing empathic responses, prompting users to share experiences, and creating a sense of shared memory by referencing well-known cultural elements, such as "I remember songs from that era by Fong Fei-fei."



Figure 4. 4 stages of objective in intent-driven conversation

The design should ensure that the robot can establish trust with users, especially when interacting with elderly individuals. Incorporating pre-and post-tests for trust levels and considering user anxiety are essential for measuring the effectiveness of the interaction. Research should focus on identifying conversational patterns that resonate with users, such as topics that elicit more engagement and discussion. Additionally, the robot should be designed to initiate small talk, use back-channel cues (like "mm-hmm" and "uh-huh") to fill pauses, and adjust its speech rate to match the user's pace.

Personalization goes beyond merely character traits; it should include adapting the robot's role based on the user's identity and preferences. For example, the robot might switch roles between a peer and a grandchild depending on the context. Early conversations should gather basic information about the user, such as their background, to facilitate more personalized interactions in subsequent engagements.

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